On the Optimality of Global Improvement Regularization over Local KL Maximization in Incremental Direct Preference Optimization

Ayosh Singh Yadav(B.Tech IIT Delhi)

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Abstract

Direct Preference Optimization (DPO) aligns generative models by reparameterizing the Kullback–Leibler (KL)-regularized reward maximization objective directly in terms of policy probability ratios. While standard DPO assumes a static offline preference dataset and a fixed inverse-temperature parameter β, practical post-training environments increasingly operate in dataset-free, incremental regimes where candidate pairs are sampled on-policy and evaluated sequentially by an automated reward judge. In this paper, we investigate the theoretical selection of the KL regularization trajectory under incremental preference generation. We formulate and compare two competing hypotheses: (1) a *Local-Maximum KL Strategy* ($\mathcal{H}_1$) that seeks to maximize step-wise distributional shift within a local neighborhood, and (2) a *Global-Maximum Net Improvement Strategy* ($\mathcal{H}_2$) that computes an analytical regularization parameter β^* maximizing expected global policy improvement penalized by finite-sample estimation variance. Using Bauer's Maximum Principle and second-order Fisher information analysis, we prove that $\mathcal{H}_1$ is ill-posed on the interior of the probability simplex and induces boundary mode collapse. Conversely, via a second-order thermodynamic cumulant expansion of the log-partition function, we prove that $\mathcal{H}_2$ admits a unique interior global maximizer $\beta^* = (\lambda\bar{\sigma}_r^2/2)^{1/3}$ and achieves $\mathcal{O}(\sqrt{T})$ cumulative regret. Finally, we present an online primal-dual calibration algorithm for incremental, dataset-free DPO alignment.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) and its closed-form counterpart, Direct Preference Optimization (DPO) [1], rely on reverse Kullback–Leibler (KL) divergence regularization to prevent a parametric policy π_θ from degenerating as it optimizes preference signals relative to a reference policy π_{ref}. Formally, given a prompt distribution $x \sim \mathcal{P}$ and a latent reward function $r(x, y)$ defined over discrete output sequences $y \in \mathcal{Y}$, the canonical KL-constrained optimization problem is given by:

$$\max_{\pi_\theta \in \Delta^{|\mathcal{Y}|-1}} \mathcal{J}(\pi_\theta) = \mathbb{E}_{x\sim\mathcal{P},\, y\sim\pi_\theta(\cdot|x)}\big[r(x,y)\big] - \beta\, \mathbb{D}_{\text{KL}}\big(\pi_\theta(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x)\big), \tag{1}$$

where $\beta > 0$ controls the strength of the KL penalty. Exploiting the exact analytical solution to Eq. (1), Rafailov et al. [1] demonstrated that the ground-truth reward can be expressed up to a prompt-dependent partition function $Z(x)$ as $r(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x)$. Substituting this reparameterization into the Bradley–Terry preference model yields the standard DPO loss over preference triplets (x, y_w, y_l):

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_w,y_l)}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right], \tag{2}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic sigmoid function.

Despite its empirical success on static datasets, DPO exhibits severe sensitivity to the choice of β and to off-policy distributional drift [2, 4]. When no pre-existing preference dataset is available, alignment must be performed incrementally via Online AI Feedback (OAIF) [3]: at each step t, the current policy π_{θ_t} generates candidate responses $(y_t^{(1)}, y_t^{(2)})$, an evaluator assigns preference labels $(y_{w,t}, y_{l,t})$, and the policy executes an immediate parameter update. In this incremental regime, improper control of the KL divergence either amplifies single-sample annotation noise into catastrophic policy collapse or stalls learning altogether.

In this work, we provide a rigorous mathematical comparison between targeting a local maximum of the step-wise KL divergence versus computing an optimal β^* that maximizes global expected policy improvement.

2 Formal Hypotheses

Let $\Delta^{|\mathcal{Y}|-1} = \{\pi(\cdot \mid x) \in \mathbb{R}^{|\mathcal{Y}|} : \pi(y \mid x) \geq 0, \sum_{y\in\mathcal{Y}} \pi(y \mid x) = 1\}$ denote the conditional probability simplex over finite vocabulary/token sequences $\mathcal{Y}$, and let $\text{int}(\Delta^{|\mathcal{Y}|-1})$ denote its strict interior where $\pi(y \mid x) > 0$ for all $y \in \mathcal{Y}$. We evaluate two formal hypotheses regarding KL regularization in DPO:

Hypothesis 1 (Local-Maximum KL Strategy ($\mathcal{H}_1$))**.** *At each incremental step t with prompt x_t, selecting the regularization parameter and step trajectory to converge toward a local maximum of the step-wise KL divergence functional*

$$K(\theta; x_t) \triangleq \mathbb{D}_{\text{KL}}\big(\pi_\theta(\cdot \mid x_t) \parallel \pi_{\text{ref}}(\cdot \mid x_t)\big) \tag{3}$$

within a neighborhood $\mathcal{N}(\theta_t)$ maximizes the directional separation between preferred response $y_{w,t}$ and rejected response $y_{l,t}$ while preserving stable gradient updates.

Hypothesis 2 (Global-Maximum Net Improvement Strategy ($\mathcal{H}_2$))**.** *Computing the regularization parameter β^* to maximize the global expected net policy improvement functional over the prompt distribution $\mathcal{P}$—accounting for finite-sample quadratic estimation drift—yields a uniquely well-posed interior solution $\beta^* \in (0, \infty)$ that guarantees bounded policy divergence and sublinear cumulative regret $\mathcal{O}(\sqrt{T})$ under incremental preference generation.*

3 Mathematical Derivations and Proofs

3.1 Analysis of the Local-Maximum KL Strategy ($\mathcal{H}_1$)

We first analyze the geometric and second-order variational structure of the KL divergence functional $K(\pi) = \sum_{y\in\mathcal{Y}} \pi(y \mid x_t) \log \frac{\pi(y|x_t)}{\pi_{\text{ref}}(y|x_t)}$ on the probability simplex.

Theorem 3.1 (Non-Existence of Interior Local Maxima and Boundary Collapse under $\mathcal{H}_1$)**.** *Assume Supp$(\pi_{\text{ref}}(\cdot \mid x_t)) = \mathcal{Y}$. Then:*

1. *The functional $K(\pi)$ possesses no interior local maximum on $\text{int}(\Delta^{|\mathcal{Y}|-1})$.*
2. *Any constrained local maximum of $K(\pi)$ over the compact simplex $\Delta^{|\mathcal{Y}|-1}$ lies exclusively on the boundary $\partial\Delta^{|\mathcal{Y}|-1}$ (specifically at degenerate Dirac vertices δ_{y,y^*}), corresponding to the singular limit $\beta \to 0^+$ in DPO where log-probability ratios diverge to $\pm\infty$.*

Proof. To establish necessary and sufficient conditions for an interior local maximum of $K(\pi)$ subject to the normalization constraint $\sum_{y\in\mathcal{Y}} \pi(y \mid x_t) = 1$, we form the Lagrangian $\tilde{K}(\pi, \nu) = K(\pi) + \nu\big(1 - \sum_{y\in\mathcal{Y}} \pi(y \mid x_t)\big)$. The first-order stationary condition requires:

$$\frac{\partial \tilde{K}}{\partial \pi(y \mid x_t)} = \log \frac{\pi(y \mid x_t)}{\pi_{\text{ref}}(y \mid x_t)} + 1 - \nu = 0 \implies \pi(y \mid x_t) = \pi_{\text{ref}}(y \mid x_t) e^{\nu-1}. \tag{4}$$

Enforcing $\sum_{y\in\mathcal{Y}} \pi(y \mid x_t) = 1$ yields $\nu = 1$, establishing that $\pi = \pi_{\text{ref}}$ is the unique interior stationary point, at which $K(\pi_{\text{ref}}) = 0$.

Next, we compute the Hessian matrix of $K(\pi)$ with respect to the distribution coordinates on $\text{int}(\Delta^{|\mathcal{Y}|-1})$:

$$H_{y,y'}(\pi) = \frac{\partial^2 K(\pi)}{\partial\pi(y \mid x_t)\partial\pi(y' \mid x_t)} = \frac{\delta_{y,y'}}{\pi(y \mid x_t)}, \tag{5}$$

where $\delta_{y,y'}$ is the Kronecker delta. For any non-zero tangent vector $v \in T_\pi\Delta^{|\mathcal{Y}|-1}$ satisfying $\sum_y v_y = 0$, the quadratic form is strictly positive:

$$v^\top H(\pi)v = \sum_{y\in\mathcal{Y}} \frac{v_y^2}{\pi(y \mid x_t)} > 0. \tag{6}$$

Hence, $H(\pi) \succ 0$ everywhere on $\text{int}(\Delta^{|\mathcal{Y}|-1})$, violating the second-order necessary condition for an interior local maximum ($H(\pi) \preceq 0$). Similarly, pulling $K(\pi)$ back to parameter space θ at $\pi_\theta = \pi_{\text{ref}}$ yields the Fisher Information Matrix $\nabla^2_\theta K(\theta)\big|_{\pi_\theta=\pi_{\text{ref}}} = \mathcal{I}(\theta) \succeq 0$.

By Bauer's Maximum Principle, a continuous strictly convex function on a compact convex set attains its maxima exclusivamente at the extreme points (vertices) of the set. The extreme points of $\Delta^{|\mathcal{Y}|-1}$ are the deterministic indicator distributions $\pi(y \mid x_t) = \delta_{y,y^*}$. Under the DPO parameterization $\pi^*_\beta(y \mid x_t) = Z_\beta(x_t)^{-1}\pi_{\text{ref}}(y \mid x_t)\exp(r(x_t, y)/\beta)$ [1], converging to a vertex δ_{y,y^*} requires taking $\beta \to 0^+$. As shown by Azar et al. [2], on finite empirical preference samples $(y_{w,t}, y_{l,t})$, driving $\pi_\theta(y_{l,t} \mid x_t) \to 0$ causes the DPO implicit reward margin $\log \frac{\pi_\theta(y_{w,t}|x_t)}{\pi_\theta(y_{l,t}|x_t)}$ to diverge to $+\infty$, resulting in catastrophic entropy collapse. This refutes Hypothesis 1. □

3.2 Derivation of the Global-Maximum Improvement β^* Strategy ($\mathcal{H}_2$)

We now derive the closed-form optimal regularization parameter β^* under Hypothesis 2. Define the expected raw policy improvement over π_{ref} as:

$$\Delta(\pi) \triangleq \mathbb{E}_{x\sim\mathcal{P}}\Big[\mathbb{E}_{y\sim\pi(\cdot|x)}[r(x, y)] - \mathbb{E}_{y\sim\pi_{\text{ref}}(\cdot|x)}[r(x, y)]\Big]. \tag{7}$$

In an incremental learning setting where preferences are estimated from noisy pairwise comparisons, large policy shifts away from π_{ref} incur quadratic out-of-distribution estimation degradation governed by penalty coefficient $\lambda > 0$. We therefore define the net global improvement functional:

$$\mathcal{U}(\pi) \triangleq \Delta(\pi) - \frac{\lambda}{2}\Big(\mathbb{E}_{x\sim\mathcal{P}}\big[\mathbb{D}_{\text{KL}}(\pi(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x))\big]\Big)^2. \tag{8}$$

Theorem 3.2 (Uniqueness and Closed-Form Optimality of β^* under $\mathcal{H}_2$)**.** *Let $\mu_r(x) = \mathbb{E}_{y\sim\pi_{\text{ref}}}[r(x, y)]$ and $\sigma_r^2(x) = \text{Var}_{y\sim\pi_{\text{ref}}}(r(x, y)) > 0$ be the mean and variance of the latent reward under π_{ref}, with global average variance $\bar{\sigma}_r^2 = \mathbb{E}_{x\sim\mathcal{P}}[\sigma_r^2(x)]$. Under the second-order cumulant expansion of the DPO optimal policy family π^*_β, the global net improvement $\mathcal{U}(\beta)$ is strictly concave in the inverse temperature $\tau = \beta^{-1}$ and attains a unique global maximum at:*

$$\beta^* = \left(\frac{\lambda\bar{\sigma}_r^2}{2}\right)^{1/3}, \qquad \textit{or equivalently} \qquad \beta^* = \sqrt{\frac{\bar{\sigma}_r^2}{2\epsilon}} \tag{9}$$

when parameterized by a target global KL divergence trust region $\mathbb{E}_x[\mathbb{D}_{\text{KL}}(\pi^*_\beta \parallel \pi_{\text{ref}})] \le \epsilon$.

Proof. For any fixed $\beta > 0$, the DPO optimal policy is the Gibbs distribution $\pi^*_\beta(y \mid x) = \frac{1}{Z_\beta(x)}\pi_{\text{ref}}(y \mid x)\exp\big(\tau r(x, y)\big)$, where $\tau = \beta^{-1}$ and the log-partition function (cumulant generating function) is:

$$\Psi(\tau; x) \triangleq \log Z_\beta(x) = \log \mathbb{E}_{y\sim\pi_{\text{ref}}(\cdot|x)}\big[\exp(\tau r(x, y))\big]. \tag{10}$$

Expanding $\Psi(\tau; x)$ in a second-order Taylor (cumulant) series around the reference policy ($\tau = 0$) gives:

$$\Psi(\tau; x) = \tau\mu_r(x) + \frac{\tau^2}{2}\sigma_r^2(x) + \mathcal{O}(\tau^3). \tag{11}$$

By standard exponential-family identities, the expected reward under π_β^* is the first derivative of $\Psi(\tau; x)$ with respect to τ:

$$\mathbb{E}_{y\sim\pi_\beta^*}[r(x, y)] = \frac{\partial\Psi(\tau; x)}{\partial\tau} = \mu_r(x) + \tau\sigma_r^2(x) = \mu_r(x) + \frac{\sigma_r^2(x)}{\beta}. \tag{12}$$

Consequently, the expected policy improvement across $\mathcal{P}$ is:

$$\Delta(\beta) = \mathbb{E}_{x\sim\mathcal{P}}\big[\mathbb{E}_{y\sim\pi_\beta^*}[r(x, y)] - \mu_r(x)\big] = \frac{\bar{\sigma}_r^2}{\beta}. \tag{13}$$

Similarly, the expected KL divergence is given by the Legendre transform $\mathbb{D}_{\mathrm{KL}}(\pi_\beta^* \parallel \pi_{\mathrm{ref}}) = \tau\frac{\partial\Psi}{\partial\tau} - \Psi(\tau; x)$:

$$\mathbb{E}_{x\sim\mathcal{P}}\big[\mathbb{D}_{\mathrm{KL}}(\pi_\beta^*(\cdot \mid x) \parallel \pi_{\mathrm{ref}}(\cdot \mid x))\big] = \tau\big(\mu_r + \tau\bar{\sigma}_r^2\big) - \left(\tau\mu_r + \frac{\tau^2}{2}\bar{\sigma}_r^2\right) = \frac{\bar{\sigma}_r^2}{2\beta^2}. \tag{14}$$

Substituting Eqs. (13) and (14) into the global net improvement functional Eq. (8) yields:

$$\mathcal{U}(\beta) = \frac{\bar{\sigma}_r^2}{\beta} - \frac{\lambda\bar{\sigma}_r^4}{8\beta^4}. \tag{15}$$

Differentiating $\mathcal{U}(\beta)$ with respect to β and setting the derivative to zero provides the first-order optimality condition:

$$\frac{\partial\mathcal{U}(\beta)}{\partial\beta} = -\frac{\bar{\sigma}_r^2}{\beta^2} + \frac{\lambda\bar{\sigma}_r^4}{2\beta^5} = 0 \implies \beta^3 = \frac{\lambda\bar{\sigma}_r^2}{2} \implies \beta^* = \left(\frac{\lambda\bar{\sigma}_r^2}{2}\right)^{1/3}. \tag{16}$$

To verify that β^* is a strict global maximum, we evaluate the second derivative at $\beta = \beta^*$:

$$\left.\frac{\partial^2\mathcal{U}(\beta)}{\partial\beta^2}\right|_{\beta=\beta^*} = \frac{2\bar{\sigma}_r^2}{(\beta^*)^3} - \frac{5\lambda\bar{\sigma}_r^4}{2(\beta^*)^6} = \frac{2\bar{\sigma}_r^2}{(\beta^*)^3} - \frac{5\bar{\sigma}_r^2}{(\beta^*)^3} = -\frac{3\bar{\sigma}_r^2}{(\beta^*)^3} < 0. \tag{17}$$

Moreover, expressed in terms of $\tau = \beta^{-1} \in (0, \infty)$, $\tilde{\mathcal{U}}(\tau) = \bar{\sigma}_r^2\tau - \frac{\lambda}{8}\bar{\sigma}_r^4\tau^4$ satisfies $\tilde{\mathcal{U}}''(\tau) = -\frac{3}{2}\lambda\bar{\sigma}_r^4\tau^2 < 0$ everywhere on $(0, \infty)$, proving that β^* is the unique global maximizer. Finally, equating the KL divergence in Eq. (14) to a target trust-region bound $\epsilon > 0$ yields $\frac{\bar{\sigma}_r^2}{2(\beta^*)^2} = \epsilon \implies \beta^* = \sqrt{\frac{\bar{\sigma}_r^2}{2\epsilon}}$. This establishes Hypothesis 2. □

4 Theoretical Comparison

Table 1 summarizes the analytical properties of $\mathcal{H}_1$ and $\mathcal{H}_2$. In the incremental regime where preference triplets $(x_t, y_{w,t}, y_{l,t})$ arrive sequentially for $t = 1, \ldots, T$, let $g_t(\theta_t) = \nabla_\theta\mathcal{L}_{\mathrm{DPO}}^{(t)}(\theta_t)$ denote the stochastic gradient. Assuming bounded score gradients $\|\nabla_\theta \log \pi_\theta(y \mid x)\| \le G$, the step-wise gradient norm is bounded by $\|g_t(\theta_t)\| \le 2\beta_t G$.

Under $\mathcal{H}_2$, maintaining β_t within a compact interval $[\beta_{\min}, \beta_{\max}]$ around β^* via online dual ascent with step size $\eta_t = \mathcal{O}(1/\sqrt{t})$ guarantees that the cumulative online regret satisfies:

$$\mathrm{Regret}(T) = \sum_{t=1}^{T}\mathcal{L}_{\mathrm{DPO}}^{(t)}(\theta_t) - \min_{\theta\in\Theta_\epsilon}\sum_{t=1}^{T}\mathcal{L}_{\mathrm{DPO}}^{(t)}(\theta) \le \mathcal{O}\big(G\beta_{\max}\sqrt{T}\big), \tag{18}$$

whereas under $\mathcal{H}_1$, driving π_θ toward the boundary causes $\|\nabla_\theta \log \pi_\theta(y_{l,t} \mid x_t)\| \to \infty$ as $\pi_\theta(y_{l,t} \mid x_t) \to 0$, inducing $\mathcal{O}(T)$ linear regret and immediate overfitting to noisy single-sample comparisons [2, 4].

Table 1: Theoretical Comparison of KL-Divergence Strategies in Incremental DPO

Theoretical Property	**Local-Maximum KL ($\mathcal{H}_1$)**	**Global-Maximum Improvement β^* (**
Variational Curvature	Strictly convex ($H(\pi) \succ 0$)	Strictly concave in $\tau = \beta^{-1}$ ($\tilde{\mathcal{U}}''(\tau) < 0$)
Interior Stationary Point	Global minimum only ($\pi = \pi_{\text{ref}}$)	Unique global maximum $\beta^* = (\lambda\bar{\sigma}_r^2/2)^{1/3}$
Asymptotic Policy Support	Degenerate Dirac boundary (δ_{y,y^*})	Full-support Gibbs distribution ($\pi^*_{\beta^*}$)
Log-Ratio Variance	Unbounded ($\log \frac{\pi_\theta}{\pi_{\text{ref}}} \to \pm\infty$)	Bounded by $2\epsilon = \bar{\sigma}_r^2/(\beta^*)^2$
Incremental Regret (T steps)	$\mathcal{O}(T)$ (linear overfitting to noise)	$\mathcal{O}(\sqrt{T})$ (sublinear convergence)

5 Methodology for Incremental Dataset-Free DPO

When no static preference dataset exists ahead of training, preferences must be generated and consumed incrementally after each model response [3]. Because the true reward variance $\bar{\sigma}_r^2$ is unknown *a priori*, we estimate β^* online by combining empirical log-ratio variance tracking [4] with primal-dual KL trust-region calibration (Algorithm 1).

Algorithm 1 Incremental Dataset-Free DPO with Online Global-Improvement β_t Calibration

Require: Reference policy π_{ref}, initial policy $\pi_{\theta_1} \leftarrow \pi_{\text{ref}}$, target KL budget $\epsilon > 0$, bounds $[\beta_{\min}, \beta_{\max}]$, EMA rate $\alpha \in (0, 1)$, learning rates $\eta_\theta, \eta_\beta > 0$.

Initialize $\beta_1 \leftarrow \beta_0$, running KL estimate $\hat{D}_0 \leftarrow \epsilon$, running margin variance $\hat{V}_0 \leftarrow 1.0$.
for $t = 1, 2, \ldots, T$ **do**
  Receive prompt $x_t \sim \mathcal{P}$ and sample $K \geq 2$ on-policy responses $\{y_t^{(1)}, \ldots, y_t^{(K)}\} \sim \pi_{\theta_t}(\cdot \mid x_t)$.
  Query automated evaluator (e.g., reward judge) to rank candidates and form pair $(y_{w,t}, y_{l,t})$.
  if evaluator indicates a tie or invalid candidate **then**
    continue ▷ Skip non-decisive or corrupted incremental samples
  end if
  Compute log-ratios $u_{w,t} = \log \frac{\pi_{\theta_t}(y_{w,t}|x_t)}{\pi_{\text{ref}}(y_{w,t}|x_t)}$ and $u_{l,t} = \log \frac{\pi_{\theta_t}(y_{l,t}|x_t)}{\pi_{\text{ref}}(y_{l,t}|x_t)}$.
  Compute empirical margin $M_t = u_{w,t} - u_{l,t}$ and step KL proxy $\hat{k}_t = \frac{1}{2}(u_{w,t}^2 + u_{l,t}^2)$.
  Update EMA KL tracker: $\hat{D}_t \leftarrow (1 - \alpha)\hat{D}_{t-1} + \alpha\hat{k}_t$.
  Update primal policy parameters via DPO gradient:
    $\theta_{t+1} \leftarrow \theta_t + \eta_\theta \beta_t \sigma(-\beta_t M_t)\big(\nabla_\theta \log \pi_{\theta_t}(y_{w,t} \mid x_t) - \nabla_\theta \log \pi_{\theta_t}(y_{l,t} \mid x_t)\big)$.
  Update dual regularization parameter β_{t+1} toward global optimum β^*:
    $\beta_{t+1} \leftarrow \Pi_{[\beta_{\min}, \beta_{\max}]}\Big(\beta_t + \eta_\beta(\hat{D}_t - \epsilon)\Big)$.
end for
return Aligned policy $\pi_{\theta_{T+1}}$.

In Algorithm 1, whenever an incremental preference pair causes an excessive distributional shift ($\hat{k}_t > \epsilon$), the dual ascent step increases β_{t+1}, tightening the KL constraint on subsequent steps. Conversely, when policy updates remain overly conservative ($\hat{k}_t < \epsilon$), β_{t+1} decreases toward β^* to accelerate policy improvement.

6 Conclusion

We have proven that pursuing a local maximum of the KL divergence in DPO ($\mathcal{H}_1$) violates second-order interior optimality conditions and forces the policy toward degenerate simplex boundaries. By contrast, choosing β^* to maximize global expected policy improvement penalized by quadratic estimation drift ($\mathcal{H}_2$) yields a strictly concave objective in inverse temperature

with a unique closed-form interior solution $\beta^* = (\lambda\bar{\sigma}_r^2/2)^{1/3}$. Coupled with online primal-dual calibration, this strategy enables stable, dataset-free incremental alignment with sublinear regret.

References

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